

Institute of Actuaries of India

Subject SA4: Pension & Other Employee Benefits

May 2014 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the paper setters with the aim of helping markers of scripts so as to have a framework and be consistent while evaluating answers. The solutions given are only indicative. It is realized that there could be other points as valid answers and the marker may give credit for any such alternative approach or interpretation which the marker considers to be appropriate.

Solution 1 :**i) Executive Summary :****Introduction**

- 1.1** The objective of this report is to provide comprehensive inputs for the Company to understand the potential costs of continuing to providing the Post-Retirement Medical benefit (PRMB) on the i) Balance sheet and ii) Profit and Loss account.
- 1.2** To achieve this objective, I have carried out a projection of the estimated liability and accounting expense for the next five years. I have also carried out a review of the medical inflation assumption and have provided the sensitivity of the estimated liability and accounting expense for the next five years to this assumption. Section 2.1 of this report summarises the above said results.
- 1.3** There is an increase of x% in liability at the end of five years. If the medical inflation assumption is increased by 100 basis points, then the liability at the end of five years is y%. The liability is sensitive to the medical inflation assumption. The cost basis should be reviewed periodically in the future to ensure that the cost trends are suitably incorporated in the basis.
- 1.4** This report is addressed to the senior management of the Company. It is not a report for accounting purposes under Indian Accounting Standard 15 (Revised) or any UK or International standard. However, this report has been prepared in accordance with the Guidance Note 11 (Actuarial Investigation of Retirement Benefit Schemes) and Guidance Note 18 (Retirement Benefit Schemes – Actuarial Reports) issued by the Institute of Actuaries of India, where relevant.

Project Benefit Obligation (PBO)

The actuarial liability for the PRMB scheme as at 31 March 2014 and the estimated liability for the next five years based on current basis and sensitivity (+/- 100 basis points medical inflation rate assumption) are summarised below to showcase the cost of the scheme and the sensitivity of the liability to the medical inflation assumption.

Liability as at	Liability (INR Million) (current basis)	Liability (INR Million) (inflation +100 bp)	Liability (INR Million) (inflation -100 bp)
31 March 2014			
31 March 2015			
31 March 2016			
31 March 2017			
31 March 2018			
31 March 2019			

[5]

ii) Data required :

The following data is required as at 31 March 2014 and the projection for the next five years:

- Membership data for all eligible employees in service
 1. Employee id
 2. Date of birth
 3. Date of joining the Company
 4. Spouse date of birth
 5. Retirement age
- Membership data for all eligible employees retired
 1. Employee id
 2. Date of birth
 3. Spouse date of birth
 4. Insurance premium paid in last renewal
- Scheme data
 1. Insurance premium paid per member aged 60 (including spouse) in the last renewal and the previous two years
 2. Reasons for any material increase in the premium over the last three years (high claims for instance)
 3. Scheme provisions and any change in the current year
 4. Any past changes in the scheme provisions along with reasons
 5. Any proposed changes in the scheme provisions in the future
- Data for projections
 1. New entrant profile for the purpose of projection (average age at entry, spouse age difference)
 2. Increase in head count of eligible in service employees over the next five years

[5]

iii) Data checks :

- Membership reconciliation
 1. I have done a member reconciliation for the inter valuation period comparing the membership at the last valuation with the membership at the current valuation and the movements that took place between the two valuations. I have done this reconciliation for actives and retired members independently.
 2. I have checked for missing data to ensure that no existing members as at previous valuation have been missed out without suitable reasons for removal (death, resignation etc...)
 3. I have also checked that all new entrants during the year who joined the Company and eligible for the scheme, have been included.
 4. I have done some simple checks to verify consistency between data. For example dates of birth, dates of joining etc...
 5. I have also done some random spot checks to ensure that there are no blank fields and have obtained confirmation from the Company that the data is consistent with the payroll data.

[5]

iv) Assumptions :

The following assumptions have been used for the valuation :-

The general approach to setting the assumptions is to consider the current short-term position and then blend this with a long-term view based on past experience and general expectations to the extent required by the respective assumptions.

Financial Assumptions - Discount Rate

- 1.1** Basis: For deciding on the rate of discounting for the period up to retirement, I have used the yields on long term Indian Government bonds due to lack of deep market for corporate bonds in India

Other Financial Assumptions

- 1.2** The financial results summarised in this report involve actuarial calculations that require assumptions about future events. The assumptions used for the purpose of this report have been set based on discussions with the Company. The Company has confirmed the assumptions. I believe that the assumptions used in this report are reasonable for the purposes for which they have been used and are summarised as below. Would like to highlight that the liabilities are sensitive to all the assumptions used for valuation purposes and the change in these assumptions may have a substantial impact on the liabilities.
- 1.3** The Company has confirmed the following assumptions for a base scenario (Best estimate).

Discount Rate	8.00% p.a.
Medical Inflation Rate*	8.00% p.a.
Insurance premium at age 60 for cover of 500,000	INR 20,000

* The inflation is applicable for the insurance premiums payable in the future for providing the benefit

Demographic Assumptions

These assumptions are required to assess the timing and probability of a payment taking place. The key assumptions used for the base scenario are as set out below and have been agreed with the Company. The assumptions are consistent with prevalent practice in such an exercise considering the membership profile.

Mortality Rate (pre-retirement)	Indian Assured Lives Mortality (2006-08) Ultimate*
Mortality Rate (post-retirement)	LIC a (1996-98) Ultimate rates
Spouse Age Assumption	5 years less than the employee**
Attrition rate	7.00% p.a.
Early Retirement	No early retirement before age 60 assumed

*As published by IRDA and adopted as Standard Mortality Table as recommended by Institute of Actuaries of India effective 1 April 2013

**We have assumed that every member of the scheme has a spouse 5 years younger to the member

New entrant profile for the purpose of the projection:

Age at entry	27 years
Spouse Age Assumption	5 years less than the employee

Please note that discount rate, base insurance cost and medical inflation assumption are critical assumptions and the liabilities and projections are sensitive to these assumptions.

[5]

v) Key Assumptions and Sensitivity :

The sensitivity of the liability and the projected expense for the next five years to change in medical inflation is as below :

Liability as at	Liability & Expense (INR Million) (current basis)		Liability & Expense (INR Million) (inflation +100 bp)		Liability & Expense (INR Million) (inflation -100 bp)	
	PBO	Expense	PBO	Expense	PBO	Expense
31 March 2014						
31 March 2015						
31 March 2016						
31 March 2017						
31 March 2018						
31 March 2019						

An increase in medical inflation by 100 basis points has a _% increase in the liability in year one and a _% increase over five years.

The liability and project expense are quite sensitive to mortality improvements post retirement and a summary of the impact due to increase in longevity by one/ two years is as below.

Liability as at	Liability & Expense (INR Million) (current basis)		Liability & Expense (INR Million) (Longevity +1 year)		Liability & Expense (INR Million) (Longevity +2 years)	
	PBO	Expense	PBO	Expense	PBO	Expense
31 March 2014						
31 March 2015						
31 March 2016						
31 March 2017						

31 March 2018			
31 March 2019			

An increase in longevity by 1 year has a _% increase in the liability in year one and a _% increase over five years.

[5]

vi) Review of Medical Cost Inflation – Methodology :

- 1.4** I have used the Projected Unit Credit (PUC) actuarial method to assess the Plan's liabilities, including those related to death-in-service and incapacity benefits. Past Service liability and Total service liability has been calculated under this method.
- 1.5** Under the PUC method a "projected accrued benefit" is calculated at the beginning of the year and again at the end of the year for each benefit that will accrue for all active members of the Plan. The "projected accrued benefit" is based on the Plan's accrual formula and upon service as of the beginning or end of the year, but using a member's final compensation, projected to the age at which the employee is assumed to leave active service. The Plan liability is the actuarial present value of the "projected accrued benefits" as of the beginning of the year for active members

1. Review of Medical Cost Inflation – Assumptions :

- Data required for the review
 - a) Historical data on the actual premium increases in the past three to five years. While the past is a useful indicator, it should be recognised that the past data will also be impacted by economic changes and medical advancements. The past data need to be taken into account suitably
 - b) Insurance Company forecasts for premium cost increases
 - c) Historical market statistics (e.g. past premium cost increases)
 - d) Information from the Company on future expected claims cost increases
- The review process and key considerations
 - a) The data for medical inflation as obtained from different sources of information for the past five years has been summarised as below :

Medical inflation for the year ending	As per company experience	As per market data	As per insurance company
31 March 2010			
31 March 2011			
31 March 2012			
31 March 2013			
31 March 2014			

- i. There is a positive correlation across the three experience sets. There has been a significant increase in the premium cost over the past five years.
- ii. The medical inflation on average has been +_ basis points over and above the price inflation over the past five years.

- iii. The future trend as confirmed by the Company and the market information seem to corroborate the fact that medical inflation will continue to be high.
- iv. Based on the above information, would expect the same inflation+ difference to remain in the future.
- v. Taking into account the below aspects, would recommend that the current assumption is retained as these indicate that the past experience is likely to continue in the future. Would however recommend a review of the basis in three years.
 - ii. The Past experience and the trend in past experience
 - iii. The market data and projected estimates for increases
 - iv. Employee demographic profile and possible medical claim scenarios in the future (the Company has more older people and also a higher risk because of concentration of risks due to higher no of diabetes cases etc..)
 - v. The implications on cost due to regulatory changes expected in the future (standardisation of procedures for instance)
 - vi. The implications on cost due to medical advancements expected in the future (making it more expensive or more affordable)
 - vii. The implications on cost due to wellness measures in the Company that can prevent potential cost increases in the future

[10]

(Additionally up to 5 marks for the totality of the report)

[Total Marks-40]

Solution 2 :**i) Choices available to the company :**

The choices available to the software company for financing the benefits are:

- pay-as-you-go
- smoothed pay-as-you-go
- terminal funding
- just-in-time funding
- regular contributions
- lump sum in advance.

[3]

ii) Criteria used for choosing the funding method :

The criteria used for choosing the financing method are:

- security
- stability
- increasing age (durability)
- realism
- liquidity
- flexibility
- opportunity cost

[3]

iii) Benefits related to potential service :

The following funding methods or approaches could be applied for employee benefits related to potential service.

Uniform accrual to normal retirement

Most common method is to fund the benefit as if it accrues uniformly over the period to normal retirement. The part accrued to date would be funded in the same way as the normal retirement benefits.

Contributions could be made to meet the part of the benefit relating to service after the assumed date of leaving in one of the following ways:

- a) A terminal funding approach – fund that additional element as it actually becomes payable.
- b) A risk cost (or current cost) approach – pay contributions depending on the additional benefits that are expected to become payable over the next year and the probability of that benefit becoming payable (usually based on the valuation assumptions). If the benefits are insured, insurance companies are likely to use the risk cost approach though with their own assumptions.

Uniform accrual to date of payment

The benefit could be funded as if it accrues uniformly over the period until it becomes payable. For example, the death benefit payable at each future age would be assumed to accrue uniformly between the member's entry to the scheme and the possible year of death. This approach is required under US accounting standards.

Current cost approach for all accrual

The current cost (or risk cost) approach could be used for the full benefit (including the accrued part). This approach can cause difficulties when carrying out discontinuance checks which require the accrued part to be fully funded.

Benefits independent of service:

The current cost approach as above is the most common approach in relation to the funding of service-independent death-in-service lump sum benefits (eg four times the salary on death) where the benefit is independent of service.

[5]

iv) Taxation aspects that will impact the company's profit and loss :

If the benefits are paid as and when they arise, they would qualify as a deductible in the company's profit and loss at the time of payment.

The insurance premiums paid for securing death-in-service benefits would qualify for tax deduction.

If the benefits are funded in advance through a tax approved trust, then the contribution from the employer towards the fund would qualify as a deductible. The main conditions include that the Scheme should be run by the trustees, the trust should be irrevocable in nature, and the employer should be contributor to the fund.

However there could be a cap on such a deductible as per the various legislations (this is 8.33% for gratuity and 27 % for provident fund and superannuation put together). Any excess contribution over the cap would not be tax deductible.

The past service contributions made by the employer could be spread for tax deduction over a period of maximum five years.

If the benefits are funded in advance through a tax approved trust, the investment income (both interest and capital gains) is not taxable.

Thus the excess outgo for the employees in terms of the better benefit structure may qualify for tax deduction. However the extent of such benefit depends on the various limits as applicable as mentioned above.

[5]

v) Taxation aspects that will impact the employee :

The insurance premium paid by the employer for securing death-in-service benefits would not be considered as a perquisite in the hands of the employees.

If the benefits are funded in advance through a tax approved trust, then the contribution from the employer towards the fund would not be considered as a perquisite in the hands of the employees. However for Superannuation scheme any contribution by employer in excess of Rs.1,00,000 would be considered as a perquisite in the hands of the employee.

The taxation on the benefits are as below:

- Any death benefits would not be taxed in the hands of the employee / beneficiaries
- Any gratuity benefits as computed as per Payment of Gratuity Act, 1972 subject to a maximum of Rs.10,00,000 will not be taxed. Thus the additional benefit for the balance service up to normal retirement and any benefit in excess of Rs.10,00,000 is taxable in the hands of the employee. Thus the additional gratuity benefits would be taxable in the hands of the employees.
- Any superannuation benefit paid as a lump-sum is not taxable (this lump sum is restricted to a maximum of one-third in case the benefit is secured from a tax approved trust fund, any excess over one-third is taxable on receipt in the hands of the employee). This includes the additional benefits for service up to normal retirement.
- Any superannuation benefit paid as a pension is taxed as income in the hands of the employee at the time of receipt. This includes the additional benefits for service up to normal retirement.
- Any other benefits paid is taxable in the hands of the employee.

[4]

[Total Marks-20]

Solution 3 :**i) a) Reasons for the actuarial gain, variation in the funding status and the expense :**

The actuarial gain or loss could arise due to:

- *Experience different from Assumptions:* any gains or losses in the actuarial liability arising from items of the scheme experience being different from those assumed at the beginning of the year
- *Changes in Assumptions:* the change in the actuarial liability arising from changes in the financial and demographic assumptions at the end of the year.
- *Expected Return and Actual Return on Assets:* the difference between the expected and actual return on the scheme's assets

In the current situation, the actuarial gain has arisen in the liability side (actual liability lower than the expected liability) due to following reasons:

- *Discount Rate:* The discount rate has been changed from 8% pa to 9% pa. This would have resulted in the lower (than expected) actuarial liability and hence actuarial gain.
- *Salary Increase:* The actual salary increase is 1% during the last year whereas the assumed salary increase at the last valuation was 6% pa. This would have resulted in lower than expected salaries and hence reduced actuarial liability (compared to expected) creating the actuarial gain.
- *Leaving Service Rates:* The actual experience of 10% pa is higher than the assumed 2% pa. The actuarial gain (or loss) due to this would depend on the value of the benefits paid compared to the actuarial value of the liability held. As there are no benefits paid on leaving service, this would have also resulted in reduced actuarial liability (compared to expected) and hence also the actuarial gains.
- *Change in Assumptions:* It is not clear whether the valuation assumptions have also been changed as a result of the salary and leaving service experience. Any reduction in salary increase assumption and increase in leaving service assumption could also have resulted in reduced actuarial liability.

Further it is also mentioned that there is no (or little) actuarial gain in the asset side. Hence the actual assets are more or less close to the expected assets.

As a result of the above,

- In aggregate, there is an actuarial gain as a result of substantial actuarial gain from the liability side.
- The funded status has changed from underfunded in the previous valuation to almost fully funded in the current valuation – as the liability has reduced significantly from the expected whereas the assets have remained more or less as expected (further it is assumed that the employer's little contributions would have been made to compensate / match the service cost for the year).
- The actuarial gains are recognized immediately in the current accounting period as per AS 15 (revised, 2005). This would have resulted in the expense reducing by more than 25% from the previous valuation.

[10]

b) Suitability of the assumptions used and the points to be kept in mind before choosing the assumption for AS 15 (revised, 2005) :

Background

The actuarial assumptions used in a pension plan valuation are discount rate, withdrawal rate, salary escalation rate, pension escalation rate, mortality (both in service and post retirement) and investments return.

AS 15 (revised, 2005) has set out in paragraphs 78 to 91 the basis upon which the various assumptions to be used in the actuarial valuation for accounting purposes are to be determined.

The assumptions chosen for accounting purposes should be realistic (whereas the same could be conservative for funding purposes). The assumption chosen should reflect the expected future experience of the company employees.

APS 26 states that the actuarial assumptions are the enterprise's best estimates of the different variables and the responsibility for setting these financial assumptions rests with the company.

APS 26 further states that the actuary shall advise the client on the interpretation and determination of the various assumptions and comment on the suitability of the assumptions chosen by the client.

Discount Rate

Paragraph 78 of AS 15 (revised, 2005) states that the rate used to discount post-employment benefit obligations should be determined by reference to market yields at the balance sheet date on government bonds. The currency and term of the government bonds should be consistent with the currency and estimated term of the post-employment benefit obligations, in this case the pension benefits. The company should use the appropriate discount rate based on this prescribed approach. Hence the assumption of 9% should be based on the estimated term of the pension benefits as on the balance sheet date.

Salary Escalation Rate

In projecting the salary increases, the company needs to keep in mind two factors - one is inflation level leading to a general change in salary level. The other is career progression of the company employees. The company needs to combine the two and arrive at suitable salary growth levels. The current experience is at 1% pa. The company needs to review and decide whether the long term salary escalation rate assumption of 6% pa needs to be reduced for this experience.

Pension Escalation Rate

In projecting the pension increases, the company needs to keep in mind two factors – one is the inflation level and the second is the long term nature of the benefit payout. The company needs to appropriately arrive at a rate combining the two factors.

Leaving Service Rate

Leaving service rates during early durations of service and near retirement date will have a significant impact on the actuarial estimates. The future experience will also depend on general economic conditions as also the particular conditions affecting the company employee population. Further the rates could differ from company to company. Hence, the company needs to carefully choose the assumption which will reflect the future employee experience on a realistic basis – especially in the backdrop the current year's experience of 10% pa being higher than the long term assumption of 2% pa as in the last valuation.

Mortality

The *Indian Assured Lives Mortality (2006-08) Table* is the latest published mortality table available on Indian Assured Lives. This could be used for mortality in service. The *LIC (1996-98) Annuitants' Mortality Table* could be used for mortality post retirement.

The company needs to examine suitability of these rates and where necessary modify the assumption with an adjustment to the table by a rating up/down.

Return on Plan Assets

In deciding the rate of interest to be assumed, the company should keep in mind the current investment policy and the investment restrictions. The current investment return could be a useful starting guide. The company could appropriately adjust the same for economic trends and investment policy changes. The company has currently invested the money in a deposit administration contract with a leading insurer. Hence the expected rate of return could be based on the interest rate as decided by the insurer from time to time.

[12]

ii) a) Security of Member's Benefits and Suitable Safeguards :

The *funded status* is the excess of assets over the liabilities, *i.e.* any surplus or deficit in the pension scheme, calculated on the appropriate method and assumptions as per AS 15 (revised, 2005).

Employer Insolvency: The main risk that a beneficiary is exposed to is the sponsoring employer becoming insolvent. In such a scenario, the higher the funded status (or the assets), more will be the security. The extent of this security is dependent on the priority that is given to the funded status in the event of a wind-up when the sponsor is insolvent.

The security of benefits even otherwise would depend on the availability of moneys at the time the benefit payment is to be made. In this sense, the higher the funded status (or the assets), more will be the security.

The method and the assumptions used for an accounting valuation determine the pace of expensing. However the actual cost / benefit depend on the actual outcomes of the various assumptions. In this sense, the funded status though does not completely reflect the security

of the scheme. However if these assumptions are the best estimate of the future experience, then a positive funded status would always be better.

Scheme Related Events: Any events which are not included in the scheme rules can be considered as scheme related event. This could be payment of benefits of more favourable terms than those under the scheme rules without additional funding. This could be making or accepting a transfer payment which is different from its actuarially equivalent value. These events could affect the security of the benefits.

Poor Investment Performance: This is a manufacturing company and so its business may be cyclical. Therefore, there is a risk that the poor investment performance and requests for additional contributions occur at a time when the company's business is performing poorly and so it is least able to afford to pay additional contribution. Hence it may affect the security of the scheme.

Safeguarding Measures: The measures that could be considered include:

- *Minimum Funding:* A funding objective and a minimum funding could be stipulated to enhance the security of the benefits.
- *Investment Strategy:* The trustees could look at formulating and reviewing a suitable investment strategy for better investment management.
- *Investment Restrictions:* A restriction on the investment – in terms of self investment and risky investments would enhance the security of the benefits.
- *Scheme Regulations:* The company could influence the government or join hands with other manufacturing companies for creating a regulatory mechanism to oversee the functions of the trust and to stipulate various measures.
- *Disclosure to Members:* The company should disclose to the members information about the scheme on a regular basis including the funding level and investment performance.
- *Protection Funds / Insurance:* The company could buy insurance against investment performance or insolvency. This could be an industry organized mechanism or could be stipulated by the government.
- *Notification of Events to the Regulator (assuming regulation in force):* The events relating to the scheme could be informed to the regulators for better management of the security and expectations of the scheme members.
- *Sponsor Covenant:* The trustees could define methods for measuring and monitoring sponsor covenant and taking appropriate action.

[8]

b) Suitability of Investment Approaches :

Under AS 15 (revised, 2005) actuarial gains and losses are recognised immediately. Unless the assets are directly matched to the liabilities, there may be considerable volatility in the value of the assets relative to the liabilities.

In our context, the value of liabilities depends on the yield on government securities on the balance sheet date and thus market related.

The scheme is funded through a deposit administration type of scheme. The fund is credited with declared interest every year and the fair value of assets is set as the balance in the fund as at the particular date.

Thus value of the assets is the capital value of the deposit fund and the interest additions. The market value of the underlying investments of the insurer on the balance sheet would be different from this capital value. Thus there is a potential mismatch of assets and liabilities. This would create volatility in the funded status.

A unit linked managed fund would directly provide the market value of the underlying securities on the balance sheet date through its net asset value. Thus it could match with the liabilities and reduce this volatility in funded status.

However if there is liquidity constraint forcing realization of assets, this could create problems as the benefits may need to be paid as times of lower NAV, impacting the security of benefits. This problem is exacerbated if the scheme is poorly funded or closed for further accrual.

However the benefits can be predicted to some extent – as the retirement date and the service at retirement are predictable. This could be used to plan a proper strategy if there is a need to realize assets. However there would still be some uncertain death benefits which need to be paid.

The matching of assets and liabilities would be closer if the underlying securities are government securities of the same duration as the liabilities. Thus a G-Sec fund could provide the best possible match as for accounting is concerned.

However, a G-Sec fund may not be available. In such a case, a partial match could be obtained by investing in Debt Funds which usually could include some corporate bonds. However such funds should consider and satisfy the investment regulations for the fund.

An Equity fund will not provide sufficiently correlated assets in terms of the yield on the government securities. It may also be not possible given the investment regulations. However this could provide a good match against the salary increases.

Hence an investment with a minimum Equity Fund plus a G-Sec / Debt fund which satisfies the investment regulations and which is based on a full funding could be recommended. This approach would minimize the volatility of funded status without significantly impacting the security of the benefits.

[10]

[Total Marks-40]
